// Day 23 of 30

MOST DEVS GET THIS WRONG

SQL Query Execution Order

→ Swipe through all 6 slides

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How You Write SQL

This is NOT execution order

```
SELECT name, COUNT(*)
FROM orders
JOIN users ON orders.user_id = users.id
WHERE status = 'active'
GROUP BY name
HAVING COUNT(*) > 5
ORDER BY COUNT(*) DESC
LIMIT 10
```

💡 SQL is written in one order, executed in another

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Actual Execution Order

How the database REALLY runs it

```
1. FROM    → pick the tables
2. JOIN    → combine tables
3. WHERE   → filter rows
4. GROUP BY → group remaining rows
5. HAVING  → filter groups
6. SELECT  → pick columns
7. ORDER BY → sort results
8. LIMIT   → cut to N rows
```

💡 FROM first. SELECT is almost last!

Why This Matters

Common errors explained

```
Error: Can't use alias in WHERE

SELECT price * qty AS total
FROM orders
WHERE total > 100;  -- FAILS!

Why? WHERE runs BEFORE SELECT
→ 'total' alias doesn't exist yet

Fix: WHERE price * qty > 100
```

💡 WHERE runs before SELECT — aliases don't exist yet

WHERE vs HAVING

Now it makes sense

```
WHERE:
  → Runs BEFORE GROUP BY
  → Filters individual rows
  → Can't use aggregate functions

HAVING:
  → Runs AFTER GROUP BY
  → Filters groups
  → CAN use aggregate functions

WHERE status = 'active'  -- filter rows
HAVING COUNT(*) > 5      -- filter groups
```

💡 WHERE = before grouping. HAVING = after grouping.

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Performance Impact

Execution order affects speed

```
Bad: Filter late
  FROM orders (10M rows)
  JOIN users (expensive on 10M)
  WHERE date > '2025-01-01'

Good: Filter early
  FROM orders
  WHERE date > '2025-01-01' (filters to 1M)
  JOIN users (cheaper on 1M)

→ WHERE reduces rows BEFORE join
→ 10x faster
```

💡 Filter early, join late = fast queries

Interview Questions

You'll get asked these

```
Q: Can you use alias in WHERE?
A: No. WHERE runs before SELECT.

Q: WHERE vs HAVING?
A: WHERE filters rows, HAVING filters groups.

Q: What runs first, JOIN or WHERE?
A: JOIN first, then WHERE.

Q: How to optimize slow queries?
A: Filter early (WHERE before JOIN)
```

💡 Know execution order = ace SQL interviews

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```
done = true
```

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